

Chapter 2 : PDE with explicit solution / formulation

Sec 2.1 : Transport Eq.

$u_t + b \nabla u = 0$, $u = u(t, x)$, $b = b(t, x) \rightarrow$ 某种速度场 , u 某种属性, 在被速度场 b 搬运下不变

特征线法 : $u(t, X(t))$, $du = u_t + \frac{dx}{dt} \cdot \nabla u$ 令 $\frac{dx}{dt} = b(t, X(t))$

若 b 常, 令 $X(t) = x_0 + bt \Rightarrow u(t, X(t)) = \text{const} \Rightarrow u(t, x) = u(0, x - bt) = u_0(x - bt)$

More generally :

$$a(x, y) u_x + b(x, y) u_y = 0$$

Reparameterized as $(x(s), y(s))$, we want $x'(s) = a(x(s), y(s))$. $y'(s) = b(x(s), y(s))$

then $z(s) = u(x(s), y(s))$, $z'(s) = 0$

Note : $\frac{dy}{dx} = \frac{b(x, y)}{a(x, y)} \Rightarrow$ Solve 1st order ODE!

$$\text{Ex : } u_x + y u_y = 0$$

$$\text{Sol } \frac{dy}{dx} = \frac{y}{1} \Rightarrow y = Ce^x \Rightarrow u(x, y(x)) = \text{const} = u(0, y(0)) = u(0, c) = g(ye^{-x})$$

$$\text{Ex : } u_x + \partial_x(cu) = 0$$

$$u_x + c \partial_x u + c' u = 0 \quad \text{First : } u_t + c \partial_x u = 0 \Rightarrow \frac{dx}{dt} = c(x) \Rightarrow \text{get a solution } x = x(t)$$

$$\Rightarrow \frac{d}{dt} \underbrace{u(t, x(t))}_{z(t)} = u_t + \partial_x \cdot x'(t) = u_t + c u_x = -c' u = -c'(x(t)) \cdot u(t, x(t))$$

$$\text{Let } \hat{z}(t) = c(x(t)) z(t) \Rightarrow [\hat{z}(t)]' = c'(x(t)) x'(t) z(t) + c(x(t)) \cdot \overset{x'(t)=c(t)}{z'(t)} = 0.$$

Sec 2.2 Laplace Eq.

$$\Delta u = 0 \quad (\text{Laplace Eq.}) \quad -\Delta u = f \quad (\text{Poisson Eq.})$$

Physical Motivation:

① Geometric model (elasticity)

$\Omega \subseteq \mathbb{R}^2$ open, a curve in $\partial\Omega$, $u(x,y): \Omega \rightarrow \mathbb{R}$

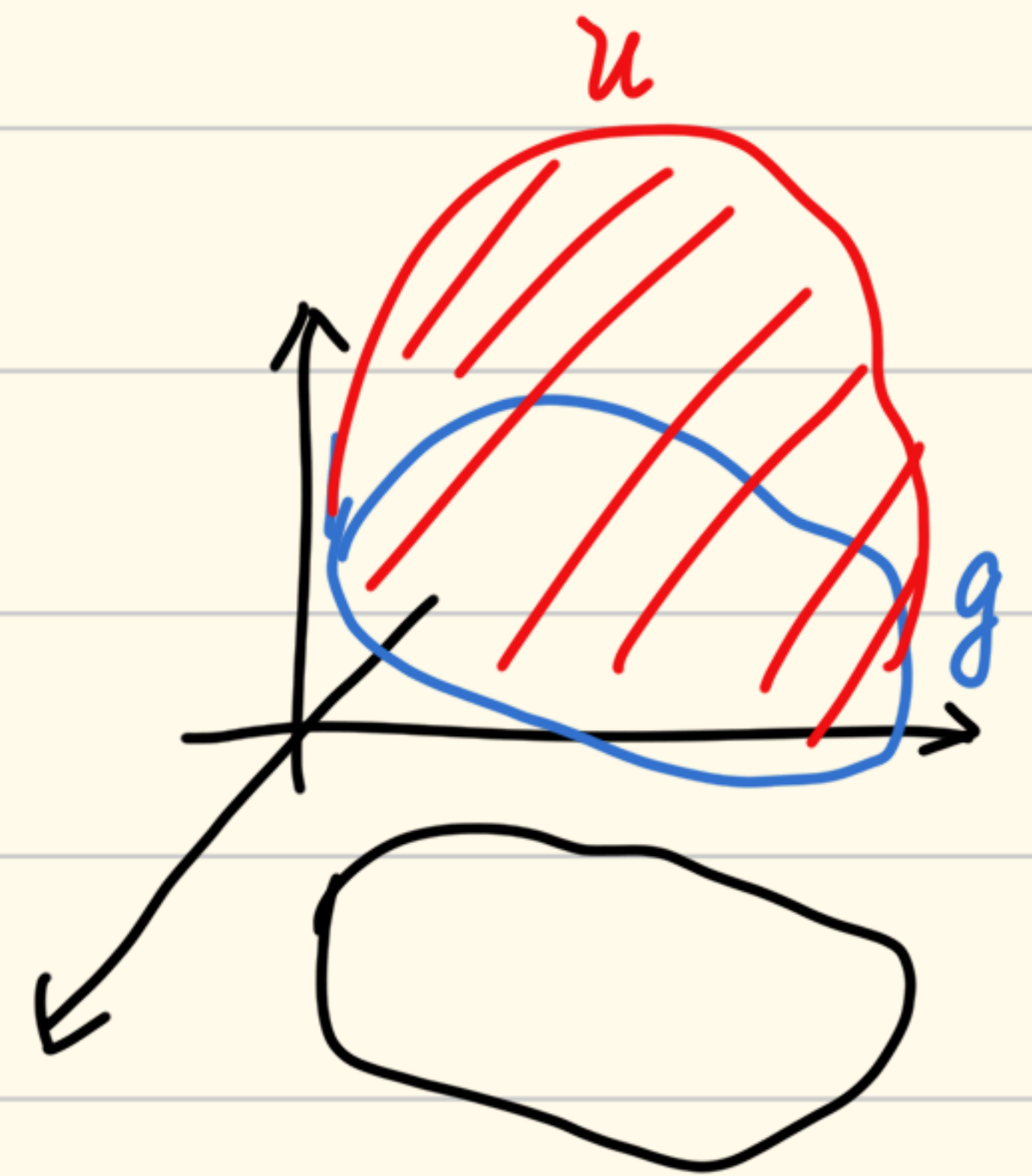
then we get an area of $\{(x,y, u(x,y)) \mid (x,y) \in \Omega\} := J(u)$

$$J(u) = \iint_{\Omega} \sqrt{1 + |Du|^2} \, dx \, dy, \quad M_g = \{u \in C^1(\Omega) \mid u|_{\partial\Omega} = g\}$$

Plateau's Prob.: Find $\min_{v \in M_g} J(v) := J(u)$

Variant of Plateau's Prob: If $|Du| < 1$, $\sqrt{1 + |Du|^2} \approx 1 + \frac{1}{2} |Du|^2$

$$F(u) = \frac{1}{2} \iint_{\Omega} |Du|^2 \, dx \, dy, \quad F(u_0) = \min_{u \in M_g} F(u)$$



Consider $\varphi \in C_0^2(\Omega) = \{\varphi \in C^2(\Omega), \varphi|_{\partial\Omega} = 0\}$, then if $u \in M_g \Rightarrow u + \varepsilon\varphi \in M_g$

let $f(\varepsilon) = F(u + \varepsilon\varphi)$, then $f(\varepsilon) \geq f(0) \quad \forall \varepsilon \in \mathbb{R}$

hence $f'(0) = 0 \Rightarrow f'(0) = \iint_{\Omega} Du_0 D\varphi \, dx dy = 0$

$$\left(\text{Green's Formula: } \iint_{\Omega} \nabla u \cdot \nabla v \, dx = \iint_{\Omega} \operatorname{div}(v \nabla u) - v \Delta u \, dx \right. \\ \left. = \iint_{\partial\Omega} v \frac{\partial u}{\partial n} \, dS - \iint_{\Omega} v \Delta u \, dx \right)$$

$$\Rightarrow f'(0) \stackrel{\text{Green}}{=} \iint_{\partial\Omega} \varphi \cdot \frac{\partial u_0}{\partial n} \, dS - \iint_{\Omega} \varphi \Delta u_0 \, dx \\ = - \iint_{\Omega} \varphi \Delta u_0 \, dx \Rightarrow -\Delta u_0 = 0 \quad (\text{Laplace Eq.})$$

Rmk: There are two boundaries condition

Dirichlet B.C.: $u|_{\partial\Omega} = g$

Neumann B.C.: $\frac{\partial u}{\partial n}|_{\partial\Omega} = g$

} Robin B.C.: $\frac{\partial u}{\partial n} + a(x)u|_{\partial\Omega} = g$

We next study properties of harmonic function i.e. $u \in C^2(\Omega)$, $\Delta u = 0$

If u is a polynomial, then we call it a harmonic polynomial

Exp: $u=1$, $u = \sum_{k=1}^n C_k x_k + a$, $u = x_1^2 - x_2^2$, $u = \operatorname{Re} f(x+iy)$ if f is analytic on $\mathbb{C}$

① Mean value Formula

$$u(x) = \int_{\partial B(x,r)} u(y) dS(y) = \int_{B(x,r)} u(y) dy$$

$$\text{Here: } \int_{\partial B(x,r)} u(y) dS(y) = \frac{1}{n\alpha(n)r^{n-1}} \int_{\partial B(x,r)} u(y) dS(y), \quad \int_{B(x,r)} u(y) dy = \frac{1}{\alpha(n)r^n} \int_{B(x,r)} u(y) dy$$

$$\begin{aligned} \text{Pf: Consider } \phi(r) &= \int_{\partial B(x,r)} u(y) dS(y) = \frac{1}{n\alpha(n)r^{n-1}} \int_{\partial B(x,r)} u(y) dS(y) \\ &= \frac{1}{n\alpha(n)} \int_{\partial B(0,1)} u(x+rz) dS(z) \end{aligned}$$

$$\phi'(r) = \frac{1}{n\alpha(n)} \int_{\partial B(0,1)} \nabla u(x+rz) \cdot z dS(z)$$

$$\stackrel{\text{Div.}}{=} \frac{1}{n\alpha(n)} \int_{B(0,1)} \operatorname{div} u(x+rz) dz = \frac{1}{n\alpha(n)} \int_{B(0,1)} r \Delta u(x+rz) dz = 0$$

$$\Rightarrow \phi(r) \equiv \text{const.} \Rightarrow \phi(r) = \phi(0) = u(x)$$

$$\text{Now: } \int_{B(x,r)} u(y) dy = \int_0^r \int_{\partial B(x,\theta)} u(y) dS(\theta) d\theta = \int_0^r \cdot u(x) n \alpha(n) \theta^{n-1} d\theta = u(x) \cdot \alpha(n) \cdot r^n$$

$$\Rightarrow \oint_{B(x,r)} u(y) dy = u(x) \quad \#$$

Conversely, we get Lemma 1:

Lemma 1: If $u \in C^2(\Omega)$ and satisfied the mean value formula, then u is harmonic.

pf: Use the notation of ϕ , we know $\phi'(r) = 0$

$$\text{Since } \forall x, r, \quad \frac{1}{n\alpha(n)} \int_{B(0,1)} \Delta u(x+rz) dz = 0 \Rightarrow \Delta u = 0$$

Lemma 2: If $u \in C(\Omega)$ and $\forall B(x,r) \subset \Omega$, $u(x) = \oint_{\partial B(x,r)} u(y) dS(y)$ then $u \in C^\infty(\Omega)$ and u is harmonic

$$\text{Mollifier: } \eta(x) = \begin{cases} C e^{-\frac{1}{1-|x|^2}} & |x| < 1 \\ 0 & |x| \geq 1 \end{cases}, \quad \int_{\mathbb{R}^n} \eta(x) dx = 1,$$

$$\eta_\varepsilon(x) = \frac{1}{\varepsilon^n} \eta\left(\frac{x}{\varepsilon}\right), \quad \text{supp } \eta_\varepsilon(x) = \overline{B(0,\varepsilon)}, \quad \int \eta_\varepsilon(x) dx = 1$$

Proof of Lemma 2: WLOG assume Ω is bounded

$$\text{Let } u_\varepsilon(x) = \eta_\varepsilon * u(x) = \int_{\Omega} \eta_\varepsilon(x-y) u(y) dy$$

Claim: $u_\varepsilon(x) \in C^\infty(\Omega)$ Check by DCT

$$\text{now } \forall x \in \Omega_\varepsilon, \quad u_\varepsilon(x) = \int_{B(x, \varepsilon)} \eta_\varepsilon(x-y) u(y) dy$$

$$= \int_0^\varepsilon \int_{\partial B(x, r)} \eta_\varepsilon(x-y) u(y) dS(y) dr$$

$$= \int_0^\varepsilon \eta_\varepsilon(r) \int_{\partial B(x, r)} u(y) dS(y) dr$$

$$= u(x) \int_0^\varepsilon \eta_\varepsilon(r) \cdot |\partial B(x, r)| dr = \int_{B(0, \varepsilon)} \eta_\varepsilon(x) dx = 1$$

$$= u(x) \quad \Rightarrow \quad u \in C^\infty(\Omega_\varepsilon) \quad \text{Let } \varepsilon \rightarrow 0^+ \quad \#$$

② Harnack Ineq.

Thm : $\Omega \subseteq \mathbb{R}^n$ open, V is connected compact subset of Ω , then there exists a constant $C = C(n, \Omega, V)$, s.t. $\forall$ nonnegative harmonic function u on Ω

$$\sup_V u \leq C \inf_V u$$

Rmk: Harnack Ineq. fails on Ω Ex: $\Omega = B(0,1)$, $u(x) = x_1 + 1$

Proof: Take $d = \text{dist}(\partial\Omega, V)$, $r = \frac{1}{2}d$, $\forall x, y \in V$, $|x - y| < r$, Notice $B(x, 2r) \supset B(y, r)$

$$\text{Then } u(x) = \frac{1}{\alpha(n)(2r)^n} \int_{B(x, 2r)} u(z) dz \geq \frac{1}{\alpha(n)(2r)^n} \int_{B(y, r)} u(z) dz = \frac{1}{2^n} u(y)$$

Cover V with open balls of radius r , $B_1 \cdots B_N$, $\Rightarrow \forall x, y$, $u(y) \leq u(x) \cdot 2^{nN}$

$$\Rightarrow \sup_{y \in V} u(y) \leq 2^{nN} \inf_{x \in V} u(x)$$

③ Strong Maximal Principle

Thm : Ω bounded open set in $\mathbb{R}^n$

$u \in C^2(\Omega) \cap C(\bar{\Omega})$ is harmonic on Ω , then:

a) $\max_{\bar{\Omega}} u = \max_{\partial\Omega} u$, $\min_{\bar{\Omega}} u = \min_{\partial\Omega} u$

b) If Ω is also connected, suppose we find $x_0 \in \Omega$, $u(x_0) = \max_{\bar{\Omega}} u$
then u is constant.

Pf: For b): take $B(x_0, r) \subset \Omega$, $u(x_0) = \frac{1}{\alpha(n)r^n} \int_{B(x_0, r)} u(y) dy \leq \frac{1}{\alpha(n)r^n} \int_{B(x_0, r)} u(x_0) = u(x_0)$

hence $\int_{B(x_0, r)} (u(x_0) - u(y)) dy = 0 \Rightarrow u(y) \equiv u(x_0) \quad \forall y \in B(x_0, r)$

Consider $S = \{z \in \Omega \mid u(z) = u(x_0)\}$ S open and close $\Rightarrow S = \Omega$

a): Find the connected component containing x_0

Corollary 1: Ω is bounded open set, $f \in C(\Omega)$, $g \in C(\partial\Omega)$

$$\begin{cases} -\Delta u = f & \text{in } \Omega \\ u|_{\partial\Omega} = g \end{cases} \quad (*) \Rightarrow (*) \text{ has at most one sol. } \in C^2(\Omega) \cap C(\bar{\Omega})$$

Pf: Assume we have 2 solutions $u_1, u_2 \in C^2(\Omega) \cap C(\bar{\Omega})$ the $w = u_1 - u_2$ satisfies

$$\Delta w = 0 \text{ in } \Omega, \quad w|_{\partial\Omega} = 0 \quad \Rightarrow \quad w \equiv 0 \text{ in } \Omega \text{ by Strong Maximal Principle.}$$

hence $u_1 \equiv u_2$

Corollary 2: Ω connected & bounded open

$$\begin{cases} \Delta u = 0 & \text{in } \Omega \\ u|_{\partial\Omega} = g \end{cases} \quad u \in C^2(\Omega) \cap C(\bar{\Omega})$$

If $g \geq 0$ then $u \geq 0$ in Ω

Furthermore, if we have $g(x_0) > 0$ for some $x_0 \in \partial\Omega$, then $u > 0$ in Ω .

④ Local estimate for harmonic functions

Thm: If u is harmonic in $\Omega \subseteq \mathbb{R}^n$, then $\forall B(x, r) \subset \Omega$, $\forall \alpha = (\alpha_1, \alpha_2, \dots, \alpha_n)$, $|\alpha| = \alpha_1 + \dots + \alpha_n \leq k$

We have estimate $|D^\alpha u(x)| \leq \frac{C_k}{r^{n+k}} \int_{B(x, r)} |u(y)| dy$, $C_0 = \frac{1}{\alpha(n)}$, $C_k = \frac{(n+k)^{n+k} (n+1)^k}{\alpha(n) (n+1)^{n+1}}$

Pf: We prove it inductively

$k=0$: Mean Value Formular $\Rightarrow C_0 = \frac{1}{\alpha(n)}$

$k=1$: u harmonic $\Rightarrow u_{x_i}$ also harmonic

consider $0 < s < r$: $u_{x_i}(x) = \frac{1}{\alpha(n)s^n} \int_{B(x, s)} u_{x_i}(y) dy$ Let $F = (0, \dots, \overset{i\text{-th}}{u}, 0, \dots, 0)$

$$u_{x_i} = \operatorname{div} F \Rightarrow \int_{B(x, s)} \operatorname{div} F dy = \int_{\partial B(x, s)} F \cdot \vec{n} dS(y) = \int_{\partial B(x, s)} u(y) \cdot n_i(y) dS(y)$$

$$\Rightarrow |u_{x_i}(x)| \leq \frac{1}{\alpha(n) \cdot s^n} \int_{\partial B(x, s)} |u(y)| dS(y)$$

$$\Rightarrow \int_0^r |u_{x_i}(x)| \alpha(n) \cdot s^n ds \leq \int_0^r \int_{\partial B(x, s)} |u(y)| dS(y) = \int_{B(x, r)} |u(y)| dy$$

$$\Rightarrow |u_{x_i}(x)| \cdot \frac{\alpha(n)}{n} r^{n+1} \leq \int_{B(x, r)} |u(y)| dy \Rightarrow |u_{x_i}(x)| \leq \frac{n}{\alpha(n)} \cdot \frac{1}{r^{n+1}} \int_{B(x, r)} |u(y)| dy$$

$$\Rightarrow C_1 = \frac{n}{\alpha(n)}$$

Suppose we have estimate for $|\alpha| \leq k-1$, consider $|\alpha| = k$, suppose $D^\alpha u = (D^\beta u)_{x_i}$ $|\beta| \leq k-1$

we consider ball $B(x, tr)$ $t \in (0, 1)$

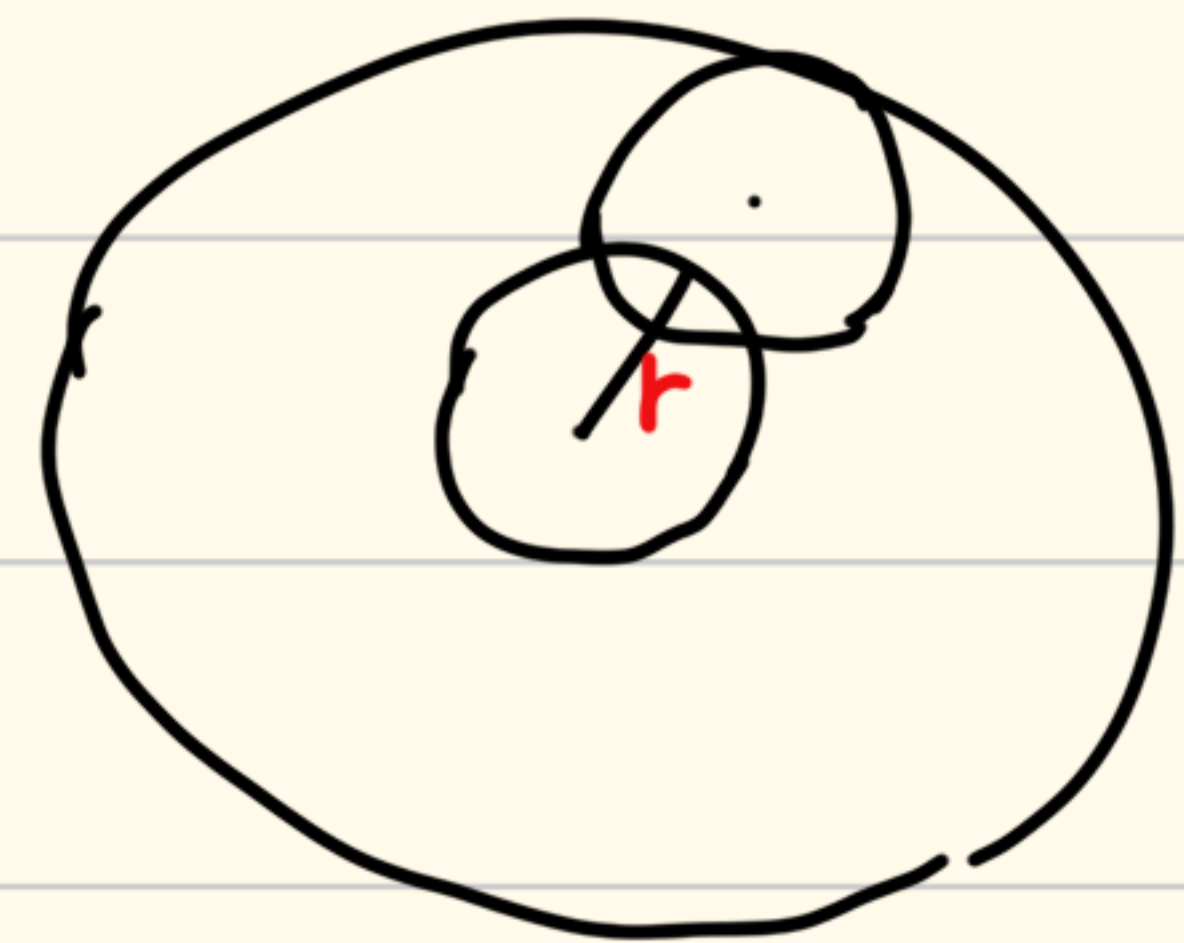
Repeat previous argument: $|D^\alpha u(x)| = |(D^\beta u)_{x_i}(x)| \leq \frac{n}{\alpha(n)} \cdot \frac{1}{(tr)^{n+1}} \int_{B(x, tr)} |D^\beta u(y)| dy$

By induction: $|D^\beta u(y)| \leq \frac{C_{k-1}}{((1-t)r)^{n+k-1}} \int_{B(y, (1-t)r)} |u(z)| dz$

$$\Rightarrow |D^\alpha u(x)| \leq \frac{1}{r^{n+k}} \frac{n}{t(1-t)^{n+k-1}} C_{k-1} \int_{B(x, r)} |u(z)| dz$$

$$\text{Let } t = \frac{1}{n+k}, \text{ we get } C_k = C_{k-1} \cdot \frac{n}{\frac{1}{n+k} \cdot (1 - \frac{1}{n+k})^{n+k-1}} = C_{k-1} \cdot \frac{(n+k)^{n+k}}{(n+k-1)^{n+k-1}} \cdot n$$

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⑤ Liouville's Thm

Thm: If u is harmonic on $\mathbb{R}^n$ & u bounded then u is constant.

Pf: $|Du| \leq \frac{C_1}{r^{n+1}} \int_{B(x,r)} |u(y)| dy$, suppose $|u| \leq k$

then $|Du| \leq \frac{C_1 k}{r^{n+1}} \cdot \alpha(n) \cdot r^n = \frac{1}{r} \cdot \tilde{C}$, let $r \rightarrow +\infty$ $\#$

⑥ Analyticity

Thm: u is harmonic in Ω then u is analytic

Pf: $\forall x \in \Omega$, $r = \frac{1}{4} \text{dist}(x, \partial\Omega)$, we want to make u as a converging series in a nbhd of x_0

$$u(x) = \sum_{k=1}^{N-1} \sum_{|\alpha|=k} \frac{D^\alpha u(x_0)}{\alpha!} (x-x_0)^\alpha + \mathcal{R}_N(x)$$

$$g(t) = u(x_0 + t(x-x_0)) \quad , \quad g(1) - g(0) = \sum_{k=1}^{N-1} \frac{g^{(k)}(0)}{k!} + \frac{g^{(N)}(t)}{N!}$$

$$g'(t) = (x-x_0)_i \partial_i u(x_0 + t(x-x_0)) = ((x-x_0) \cdot \nabla) u(x_0 + t(x-x_0))$$

$$g^{(k)}(t) = ((x-x_0) \cdot \nabla)^k u(x_0 + t(x-x_0)) = \sum_{|\alpha|=k} \frac{k!}{\alpha!} D^\alpha u(x_0 + t(x-x_0)) \cdot (x-x_0)^\alpha$$

hence we get $u(x) = \sum_{k=0}^{N-1} \sum_{|\alpha|=k} \frac{D^\alpha u(x_0)}{\alpha!} (x-x_0)^\alpha + \boxed{\sum_{|\alpha|=N} \frac{D^\alpha u(x_0 + t(x-x_0))}{\alpha!} (x-x_0)^\alpha} R_N(x)$

Recall facts : (i) $(1+\dots+1)^N = \sum_{|\alpha|=N} \frac{N!}{\alpha!}$

(ii) Stirling's Formula : $\frac{N^{N+\frac{1}{2}}}{N! e^N} \rightarrow \frac{1}{\sqrt{2\pi}}$

Estimate $R_N(x)$: From Liouville : $z = x_0 + t(x-x_0) \in B(x_0, r) \Rightarrow B(z, r) \subset B(x, 2r)$

$$|D^\alpha u(z)| \leq \frac{C_N}{r^{n+N}} \int_{B(z, r)} |u(y)| dy \leq \frac{C_N}{r^{n+N}} \int_{B(x, 2r)} |u(y)| dy$$

$$|R_N(x)| \leq \sum_{|\alpha|=N} \frac{1}{\alpha!} \cdot \frac{C_N}{r^{n+N}} \cdot C \cdot |x-x_0|^N$$

$$= \frac{n^N}{N!} \cdot \frac{C_N}{r^{n+N}} \cdot C \cdot |x-x_0|^N$$

$$(\text{take } C_N = \frac{1}{\alpha(n)} \cdot \frac{(n+N)^{n+N} \cdot (n+1)^N}{(n+1)^{n+1}})$$

$$= \frac{C}{\alpha(n)(n+1)^{n+1} \cdot r^n} \cdot \frac{(n+N)^{n+N} (n+1)^N \cdot n^N}{r^N \cdot N!} |x-x_0|^N$$

$$= \tilde{C} \cdot \frac{(n+N)^{n+N}}{N!} \cdot \left(\frac{n(n+1) |x-x_0|}{r} \right)^N$$

$$\therefore (n+N)^{n+N} = N^N \cdot \left(1 + \frac{n}{N}\right)^N \cdot (n+N)^n \sim N^N \cdot e^n \cdot (n+N)^n \sim \frac{1}{\sqrt{2\pi}} \cdot N! e^N \cdot \frac{e^n}{N^{\frac{1}{2}}} (n+N)^n$$

$$\begin{aligned} \Rightarrow |R_n(x)| &\leq \tilde{C} \cdot \frac{N! e^N e^n (n+N)^n}{N^{\frac{1}{2}}} \cdot \frac{1}{N!} \cdot \left(\frac{(n+1)n |x-x_0|}{r}\right)^N \\ &\leq \tilde{C} e^n \cdot (2N)^n \cdot \frac{e^N}{N^{\frac{1}{2}}} \cdot \left(\frac{(n+1)n |x-x_0|}{r}\right)^N \\ &\leq \tilde{C} N^{n-\frac{1}{2}} \cdot \left(\frac{e(n+1)n |x-x_0|}{r}\right)^N \end{aligned}$$

Take r_1 , $\frac{e(n+1)n r_1}{r_0} \leq \frac{1}{2}$

then $\forall x \in B(x_0, r_1) \quad |R_n(x)| \leq C \cdot N^{n-\frac{1}{2}} \cdot \left(\frac{1}{2}\right)^N \quad \text{let } N \rightarrow \infty \quad \square$

Skipped Thm 2.10, 2.11 (Try it as Ex.)

2.2.3 Solving Laplace & Poisson Eq.

Case 1 : $\Omega = \mathbb{R}^n$, $-\Delta u = 0$

Let's seek for a solution $u = u(r)$, $r = |x|$

$$u_{x_i} = u'(r) \cdot \frac{x_i}{r}, \quad u_{x_i x_i} = u''(r) \cdot \frac{x_i^2}{r^2} + u'(r) \cdot \frac{1}{r} + u'(r) \cdot x_i \cdot \frac{-x_i}{r^3}$$

$$\Delta u = u''(r) + \frac{n-1}{r} u'(r) = 0 \quad (\text{Euler Eq.})$$

$$\text{Guess: } u(r) = r^\alpha \Rightarrow \alpha(\alpha-1) + (n-1)\alpha = 0 \Rightarrow \alpha = 0 \text{ or } \alpha = -(n-2)$$

$$\alpha = 0 \rightsquigarrow u(r) = 1, \quad \alpha = -(n-2) \rightsquigarrow u(r) = r^{2-n}$$

If $n \geq 3$: we get $u(r) = a + b r^{2-n}$ } $r \neq 0!$
 $n=2$: we get $u(r) = a + b \ln r$

This tell us: there is no radially symmetric func. that satisfy $-\Delta u = 0$ globally

Def: $I(x) = \begin{cases} a_2 \ln|x| & n=2 \\ a_n |x|^{2-n} & n \geq 3 \end{cases}$ we call it **fundamental sol.** to the Laplacian Eq. a_n 待定

Prop: ① $-\Delta I(x) = 0 \quad \forall x \neq 0$

② $|\alpha| = k > 1, \quad |D^\alpha I(x)| \leq C_k \cdot |x|^{2-n-k} \quad \forall x \neq 0$

Thm: $f \in C_c^2(\mathbb{R}^n)$

Consider $u(x) = \int_{\mathbb{R}^n} I(x-y) f(y) dy$, then $-\Delta u = f$ & $u \in C^2(\mathbb{R}^n)$

Pf: Step 1: Verify $u(x)$ is finite

$$u(x) = \int_{\mathbb{R}^n} I(x-y) f(y) dy = \int_{B(x, \varepsilon)} I(x-y) f(y) dy + \int_{B(x, \varepsilon)^c} I(x-y) f(y) dy$$

$$\left| \int_{B(x, \varepsilon)} I(x-y) f(y) dy \right| \leq C \int_{B(x, \varepsilon)} |I(x-y)| dy = C \int_{B(0, \varepsilon)} |I(z)| dz = \begin{cases} \tilde{C} \int_0^\varepsilon \int_0^\varepsilon \log r \cdot r dr = C(\varepsilon - \varepsilon^2 \log \varepsilon) & n=2 \\ \tilde{C} \int_0^\varepsilon r^{2-n} \cdot r^{n-1} dr = C\varepsilon^2 & n \geq 3 \end{cases}$$

$$\left| \int_{B(x, \varepsilon)^c} I(x-y) f(y) dy \right| \leq \begin{cases} C\varepsilon^{2-n} & n \geq 3 \\ C(x) & n=2 \end{cases} \quad \#$$

Step 2: Check u is C^2

$$u(x) = \int_{\mathbb{R}^n} I(y) f(x-y) dy, \quad \frac{u(x+he_i) - u(x)}{h} = \int_{\mathbb{R}^n} I(y) \frac{f(x+he_i-y) - f(x-y)}{h} dy$$
$$= \int_{\mathbb{R}^n} I(y) \partial_i f(x-y+\theta he_i) dy, \quad 0 \leq \theta \leq 1$$

From $\partial_i f(x-y+\theta he_i) \xrightarrow{h \rightarrow 0} \partial_i f(x-y)$

and $|I(y) \partial_i f(x-y+\theta he_i)| \leq |I(y)| C \cdot \chi_{\text{紧支}}$.

Hence, from DCT: $\lim_{h \rightarrow 0} \frac{u(x+he_i) - u(x)}{h} = \lim_{h \rightarrow 0} \int_{\mathbb{R}^n} I(y) \partial_i f(x-y+\theta he_i) dy \xrightarrow{\text{DCT}} \int_{\mathbb{R}^n} I(y) \partial_i f(x-y) dy$

Similarly, $\partial_i \partial_j u = \int_{\mathbb{R}^n} I(y) \partial_i \partial_j f(x-y) dy$

Step 3: Verify $-\Delta u = f$

$$\left(\begin{array}{l} \text{Green's 2nd Identity: } \int_{\mathbb{R}^n} f \Delta g - g \Delta f \, dx = \int_{\partial \Omega} f \frac{\partial g}{\partial n} - g \frac{\partial f}{\partial n} \, dS \\ \text{pf: } \int_{\Omega} f \Delta g \, dx = \int_{\Omega} \operatorname{div}(f \nabla g) - \nabla f \cdot \nabla g \, dx = \int_{\partial \Omega} f \nabla g \cdot \vec{n} \, dS - \int_{\Omega} \nabla f \cdot \nabla g \, dx \end{array} \right)$$

Hence:

$$-\Delta u = \int_{\mathbb{R}^n} \mathcal{I}(y) (-\Delta_y) f(x-y) \, dy = \underbrace{\int_{B(x,\varepsilon)} \mathcal{I}(y) (-\Delta_x) f(x-y) \, dy}_{(I)} + \underbrace{\int_{B(x,\varepsilon)^c} \mathcal{I}(y) (-\Delta_x) f(x-y) \, dy}_{(II)}$$

$$(I) \leq C \int_{B(x,\varepsilon)} \mathcal{I}(y) \, dy \rightarrow 0 \quad (\varepsilon \rightarrow 0)$$

$$(II) = \underbrace{\int_{\partial B(x,\varepsilon)} f \frac{\partial \mathcal{I}}{\partial n}}_{II_1} - \underbrace{\mathcal{I} \frac{\partial f}{\partial n}}_{II_2} \, dS$$

$$|II_2| \leq C \int_{\partial B(0,\varepsilon)} |\mathcal{I}(y)| \, dS(y) = \begin{cases} n-2 & C\varepsilon |\log \varepsilon| \\ n \geq 3 & C\varepsilon \end{cases} \rightarrow 0$$

$|II_1|$: $n=2$ 时,

$$= f(x) (-2a_2 \alpha(2)) \rightarrow 0$$

$$II_1 = \int_{\partial B(x,\varepsilon)} f(x-y) \left(-\frac{a_r}{|y|}\right) \, dS(y) = \underbrace{\int_{\partial B^c(x,\varepsilon)} f(x) \left(-\frac{a_r}{|y|}\right) \, dS(y)}_{= f(x) (-2a_2 \alpha(2))} + \underbrace{\int_{\partial B^c(x,\varepsilon)} (f(x-y) - f(x)) \left(-\frac{a_r}{|y|}\right) \, dS(y)}_{\rightarrow 0}$$

hence: $-\Delta u \rightarrow -2a_2 \alpha(2) \cdot f(x)$ 选 $a_2 = -\frac{1}{2\pi}$

$$n \geq 3 : \quad \mathbb{I}_1 = \underbrace{\int_{\partial B(x, \varepsilon)} (f(x-y) - f(x)) \cdot a_n (n-2) \varepsilon^{1-n} dS(y)}_{\rightarrow 0} + \underbrace{\int_{\partial B(x, \varepsilon)} f(x) a_n (n-2) \varepsilon^{1-n} dS(y)}_{\rightarrow f(x) a_n (n-2) n \alpha(n)}$$

$$\Rightarrow -\Delta u = f(x) \cdot a_n (n-2) n \Rightarrow a_n = \frac{1}{\alpha(n) n (n-2)} \quad \#$$

$$\text{Summarize : } I(x) = \begin{cases} -\frac{1}{2\pi} \cdot \ln|x| & n=2 \\ \frac{1}{\alpha(n) n (n-2)} |x|^{2-n} & n \geq 3 \end{cases}$$

$$-\Delta(I * f) = f$$

Rmk: Finally introduce the "function" $\delta_0(x)$, $\int \delta_0(x) f(x) dx = f(x)$

Heuristic: $-\Delta I(x) = \delta_0(x)$, $-\Delta_x I(x-y) = 0 \quad (\forall y \neq x)$